

Jaewoo Kim

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RESEARCH INTERESTS

Space Systems Design & Operations

- Lifecycle-based space systems design considering stakeholder interactions and operational constraints
- Planning and management of space operations for efficient, resilient, and sustainable space utilization

Engineering Decision-Making Problems

- Formulation of complex engineering problems by identifying key system drivers
- Quantitative decision-making frameworks for interpretable and actionable engineering insights

EDUCATION

Korea Advanced Institute of Science and Technology (KAIST) 

Daejeon, Korea

Ph.D. in Aerospace Engineering

Feb. 2027 (Expected)

- Thesis (in progress): Maintenance Strategies for Satellite Mega-Constellations (Advisor: Prof. Jamyung Ahn )

M.S. in Aerospace Engineering

Feb. 2024

- Thesis: Optimal Satellite System Architecting Considering On-Orbit Refueling (Advisor: Prof. Jamyung Ahn )

Seoul National University (SNU) 

Seoul, Korea

B.S. in Mechanical and Aerospace Engineering

Feb. 2022

- Thesis: Celestial Navigation Using Stars and Planets on Lunar Exploration Orbit (Advisor: Prof. Changdon Kee )

PUBLICATIONS

Journal Articles (Corresponding Authors Underlined)

- [J1] **Kim, J.**, Sung, T., Hwang, W., and Ahn, J., “On-Orbit Servicing-Integrated Maintenance Strategy for Satellite Constellation,” accepted by *Journal of Spacecraft and Rockets*.
doi:10.48550/arXiv.2512.18985
- [J2] **Kim, J.**, and Ahn, J., “System Architecting for GEO Communication Satellite Considering On-Orbit Refueling,” *Journal of Spacecraft and Rockets*, Vol. 63, No. 1, 2026, pp. 176–191.
doi:10.2514/1.A36403
- [J3] **Kim, J.**, Choi, E., Ahn, J., Suh, E. S., Kim, J.-H., and Lim, D. G., “Mathematical Programming-Based Design Structure Matrix Clustering for Optimal Modular Architecture Design,” *Journal of Mechanical Design*, Vol. 147, No. 10, 2025, pp. 101401.
doi:10.1115/1.4068100
- [J4] **Kim, J.**, Ahn, J., and Sung, T., “Replenishment Strategy for Satellite Constellation with Dual Supply Modes,” *Journal of Spacecraft and Rockets*, Vol. 62, No. 5, 2025, pp. 1576–1583.
doi:10.2514/1.A36281
- [J5] Ko, J., **Kim, J.**, Choi, J., and Ahn, J., “Simultaneous Optimization of Launch Vehicle Stage and Trajectory Considering Flight-Requirement Constraints,” *International Journal of Aeronautical and Space Sciences*, Vol. 25, 2024, pp.

1563–1573.

doi:10.1007/s42405-024-00737-1

- [J6] Park, B., Howell, K. C., **Kim, J.**, and Ahn, J., “Families of Two-Impulse Optimal Rendezvous Transfers Between Elliptic Orbits,” revised manuscript under review at *The Journal of the Astronautical Sciences*.
doi:10.48550/arXiv.2603.11538
- [J7] **Kim, J.**, Song, M., Lee, J., Yu, J., Choi, E. J., Kam, H., Jo, J., and Ahn, J., “Korean Space Collision Risk Analysis Software Based on 3D-Cell Model,” under revision at *International Journal of Aeronautical and Space Sciences*.
- [J8] Choi, E., **Kim, J.**, Kim, J.-H., Suh, E. S., and Ahn, J., “Optimal Design Structure Matrix Clustering with Constraint Selections for Conflict Resolution,” under review at *Journal of Mechanical Design*.
- [J9] Lee, D., Ahn, J., and **Kim, J.**, “Multi-Domain Matrix Framework for Human Resource Decision Support,” submitted to *Systems Engineering*.

Papers Written in Korean

- [K1] **Kim, J.**, Lee, J., Choi, E. J., Choi, J., Yu, J., Jo, J., Kam, H., and Ahn, J., “Development of 3D Cell Model for Collision Risk Assessment of Space Assets,” *Journal of Space Technology and Applications*, Vol. 25, No. 2, 2025, pp. 114–124.
doi:10.52912/jsta.2025.5.2.114

Preprint & Work in Progress

- [P1] **Kim, J.**, Sung, T., Hwang, W., and Ahn, J., “Joint Replenishment Strategy for Multiple Satellite Constellations with Shared Launch Opportunities,” in preparation.
doi:10.48550/arXiv.2503.23666
- [P2] **Kim, J.**, Choi, E., Suh, E. S., Kim, J.-H., and Ahn, J., “Managing Module Boundaries Across the OEM-Supplier Interface: A Bi-Objective DSM Clustering Approach,” in preparation.
- [P3] **Kim, J.**, Lee, S., Ahn, J., Choi, E., and Kim, J.-H., “Constrained Modularity Maximization in Complex Networks via Branch-and-Price,” in preparation.
- [P4] **Kim, J.**, Ahn, J., and Choi, J., “Reinforcement Learning-Based Architecture Design Optimization of Large Space Systems with On-Orbit Servicing,” in preparation.
- [P5] **Kim, J.**, Song, M., Yu, J., Choi, E. J., Kam, H., Jo, J., and Ahn, J., “Cooperative Tasking of Space and Ground Observatories for Emergency Collision Events,” in preparation.

Proceedings of Refereed Conferences

- [C1] Lee, S., **Kim, J.**, and Ahn, J., “A Multi-Trip Time-Dependent Traveling Salesman Problem for Space Logistics,” in *2026 INFORMS Annual Meeting*, San Francisco, California, US, Nov. 1–4, 2026. (to appear)
- [C2] **Kim, J.**, Song, M., Lee, J., Yu, J., Kam, H., Choi, E.-J., Choi, J., Jo, J. H., and Ahn, J., “Inclination-Aware 2D Source-Sink Modeling for Long-Term Projection of the LEO Collision Environment,” in *27th Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference*, Maui, Hawaii, US, Sep. 15–18, 2026. (to appear)
- [C3] Song, M., **Kim, J.**, and Ahn, J., “Decision Support for Space Debris Remediation Enterprises via Two-Stage System Optimization under Various Future Scenarios,” in *27th Advanced Maui Optical and Space Surveillance Technologies (AMOS) Conference*, Maui, Hawaii, US, Sep. 15–18, 2026. (to appear)
- [C4] **Kim, J.**, Choi, J., and Ahn, J., “Flexibility-Aware Assessment and Optimization of System Architectural Decision for Large Space Systems Considering On-Orbit Servicing,” in *Asian Congress of Structural and Multidisciplinary Optimization 2026 (ACSMO 2026)*, Busan, Korea, May 17–21, 2026.
- [C5] **Kim, J.**, Sung, T., Hwang, W., and Ahn, J., “Integration of On-Orbit Servicing into Constellation Maintenance with Spare Inventory Management,” in *AIAA SciTech Forum*, Orlando, Florida, US, Jan. 12–16, 2026.
doi:10.2514/6.2026-2631

- [C6] Song, M., **Kim, J.**, Nam, Y., and Ahn, J., “Low Earth Orbit Space Debris Remediation Program for Maximizing Stakeholders’ Utility,” in *Asia-Pacific International Symposium on Aerospace Technology (APISAT) 2025*, Seoul, Korea, Oct. 27–29, 2025.
- [C7] **Kim, J.**, Sung, T., and Ahn, J., “Joint Replenishment Strategy for Multiple Satellite Constellations,” in *AIAA SciTech Forum*, Orlando, Florida, US, Jan. 6–10, 2025.
doi:10.2514/6.2025-1757
- [C8] **Kim, J.**, Lee, J., Kim, H., Choi, E. J., Choi, J., Yu, J., Jo, J., and Ahn, J., “Development of Korea Orbital Debris Evolutionary and Engineering Model,” in *75th International Astronautical Congress*, Milan, Italy, Oct. 14–18, 2024.
doi:10.52202/078360-0161
- Proceedings Written in Korean*
- [C9] **Kim, J.**, Ahn, J., Park, B., and Howell, K. C., “A Family-Based Framework for Two-Impulse Optimal Rendezvous Transfers,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Busan, Korea, Jun. 24–26, 2026.
- [C10] **Kim, J.**, Song, M., Yu, J., Kam, H., Choi, E. J., Choi, J., Jo, J. H., and Ahn, J., “3D Cell Model-Based Collision Risk Analysis for Historical and Prospective LEO Scenarios,” in *Korean Academy of Space Security (KASS) Spring Conference*, Yeosu, Korea, May 14–15, 2026.
- [C11] **Kim, J.**, Song, M., Choi, E. J., Choi, J., Yu, J., Jo, J., Kam, H., and Ahn, J., “Updates on Space Collision Risk Analysis Framework,” in *Korean Space Science Society (KSSS) Spring Conference*, Yeosu, Korea, Apr. 22–24, 2026.
- [C12] **Kim, J.**, Yi, G., and Ahn, J., “Implications of In-situ Resource Utilization from Space Logistics Perspective,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Mar. 31–Apr. 3, 2026.
- [C13] **Kim, J.**, Yi, G., and Ahn, J., “Research and Standardization of Space Logistics for Long-term Complex Space Mission,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Mar. 31–Apr. 3, 2026.
- [C14] **Kim, J.**, Sung, T., Song, M., and Ahn, J., “Integrated Inventory Model for Constellation Maintenance via Spare Satellites and On-Orbit Servicing,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Goseong, Korea, Nov. 11–14, 2025.
- [C15] **Kim, J.**, Song, M., and Ahn, J., “Optimal Design Lifetime and Propellant Loading of GEO Communication Satellites Considering On-Orbit Refueling,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Goseong, Korea, Nov. 11–14, 2025.
- [C16] **Kim, J.**, Choi, E. J., Choi, J., Yu, J., Jo, J., Kam, H., and Ahn, J., “Survey of Risk Indices for Space Objects and Implications,” in *Korean Space Science Society (KSSS) Fall Conference*, Jeju, Korea, Oct. 29–Nov. 1, 2025.
- [C17] Lee, J., Lee, J., Jang, E., Jeon, H., **Kim, J.**, Lee, C., Scheeres, D. J., and Han, J., “Asteroid Accessibility Searching Algorithm,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Yeosu, Korea, Jun. 25–27, 2025.
- [C18] **Kim, J.**, Song, M., and Ahn, J., “Survey of Research on Economic and Performance Analysis of On-Orbit Servicing and Implications,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Yeosu, Korea, Jun. 25–27, 2025.
- [C19] **Kim, J.**, Ahn, J., Choi, E. J., Choi, J., Yu, J., and Jo, J., “Development of Integrated Space Collision Risk Response System,” in *Korean Academy of Space Security (KASS) Spring Conference*, Seoul, Korea, May 9, 2025.
- [C20] **Kim, J.**, Choi, E. J., Choi, J., Yu, J., Jo, J., and Ahn, J., “Progress of Space Debris Environment and Risk Analysis Framework Development,” in *Korean Space Science Society (KSSS) Spring Conference*, Seoul, Korea, Apr. 22–24, 2025.
- [C21] **Kim, J.**, Sung, T., and Ahn, J., “Application of Space Logistics: Maintenance Strategy of Satellite Mega-Constellation,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Apr. 2–4, 2025.

- [C22] **Kim, J.**, Lee, J., Choi, E. J., Jin, C., Yu, J., Jo, J., and Ahn, J., “Development of Korean 3D Cell Model for Space Debris Environment Analysis,” in *Korean Space Science Society (KSSS) Fall Conference*, Sacheon, Korea, Oct. 28–30, 2024.
- [C23] **Kim, J.**, and Ahn, J., “An Integrated Inventory Management Model for Maintenance of Multiple Satellite Constellations,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Changwon, Korea, Jun. 26–28, 2024.
- [C24] **Kim, J.**, Ko, J., Choi, J., Ahn, J., Yoon, N., and Kim, H., “Conceptual Design of Launch Vehicle Considering Axial Acceleration Constraints,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Space Conference*, Changwon, Korea, Jun. 26–28, 2024.
- [C25] Ko, J., **Kim, J.**, Choi, J., Ahn, J., Yoon, N., and Kim, H., “Development of Conceptual Design Software for Space Launch Vehicle,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Apr. 3–5, 2024.
- [C26] **Kim, J.**, and Ahn, J., “Multiobjective Design Optimization of Commercial Satellite Considering On-Orbit Refueling Policy,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Spring Conference*, Jeju, Korea, Apr. 19–21, 2023.
- [C27] **Kim, J.**, Lee, D. U., and Ahn, J., “Research on the Overseas On-Orbit Servicing Trends and Implications,” in *Korean Society for Aeronautical and Space Sciences (KSAS) Fall Conference*, Jeju, Korea, Nov. 16–18, 2022.

RESEARCH EXPERIENCE

Graduate Research Assistant | *Strategic Aerospace Initiative, KAIST* 

Mar. 2022 – Present

- **In-Space Servicing and Manufacturing Research Center (ISMRC)**

Duration: 2025–2035; Sponsor: National Research Foundation of Korea

- Developed a maintenance framework for satellite mega-constellations integrating on-orbit servicing [C5]
- Surveyed standardization of space logistics framework [C12, C13]

- **Space Logistics Modeling and Demand Fulfillment Strategy Evaluation Framework**

Duration: 2025–2028; Sponsor: National Research Foundation of Korea

- Developed frameworks for evaluating the value of on-orbit servicing and potential architectural changes for GEO communication satellite systems [J2, C4, C15, C18]
- Developed maintenance strategies for satellite mega-constellations [J1, P1, C5, C7, C14, C21]
- Supported conceptual design for a space debris remediation program [C3, C6]
- Surveyed in-situ resource utilization from a logistics perspective [C12]

- **Multi-Criteria Decision Making for Optimal Modular System Architecture Selection**

Duration: 2024–2026; Sponsor: Hyundai Motor Company

- Surveyed multi-criteria decision-making methods and their applications in system architecting
- Developed a decision-making framework for design structure matrix-based modular architecture design considering OEM-supplier interfaces [P2]
- Supported the development of a mathematical programming approach for modular architecting with conflict resolution among design constraints [J8]
- Developed a mathematical programming-based framework for constrained modularity maximization [P3]

- **Development of an Integrated Space Risk Response System**

Duration: 2024–2027; Sponsor: Korea Astronomy and Space Science Institute

- Surveyed models for the space object environment and collision-risk analysis
- Developed a Korean orbital debris engineering model [J7, K1, C8, C10, C11, C16, C19, C20, C22]
- Developed a ground-space cooperative sensor tasking framework for operational planning [P5]

- **A Study on the Principle of Modular Architecture Engineering to Improve Level of Completion for Vehicle Architecture**

Duration: 2022–2023; Sponsor: Hyundai Motor Company

- Developed a mathematical programming approach for design structure matrix-based modular architecting [J3]
- Conducted case studies on automobile subsystems and identified improved design solutions

- **Research on ADR/OOS Applications for National Security Space Assets**

Duration: 2022; Sponsor: Korean Society for Aeronautical and Space Sciences

- Reviewed on-orbit servicing technologies and related mission concepts [C27]
- Developed concepts of operations for ADR/OOS missions

- **Development of Launch Vehicle Mission & Conceptual Design Software**

Duration: 2022–2023; Sponsor: Hanwha Aerospace

- Developed propulsion and staging modules for multidisciplinary design optimization
- Contributed to the development of an all-at-once design optimization framework considering flight requirements [J5, C24, C25]

Undergraduate Researcher | GNSS Laboratory, SNU

Mar. 2021 – Aug. 2021

- **Deep Space Navigation with Optical Sensor Data**

- Reviewed non-inertial deep-space navigation algorithms
- Evaluated the selected algorithm through numerical simulations

TEACHING EXPERIENCE

Teaching Assistant | KAIST

Sep. 2023 – Present

- AE210 Aerospace Thermodynamics – Spring 2024
- AE280 Software Application in Aerospace Engineering – Spring 2026, Spring 2025
- AE401 Aerospace System Design II – Fall 2023
- AE450 Flight Dynamics and Control – Fall 2024
- AE551 Introduction to Optimal Control – Fall 2025

Part-Time Lecturer | Humaiin Co.

Aug. 2022 – Present

- Delivered lectures and developed instructional materials covering foundational data science concepts, programming tools, and LLM-based workflow automation.
- Institutions: Busan City Government, Korea Education & Research Information Service (KERIS), Statistics Korea (KOSTAT), Ewha Womans University, Sookmyung Women's University, Seoul AI Foundation, Hanwha Solutions, Lotte Chemical

PROFESSIONAL SERVICES

- Reviewer for *INCOSE Systems Engineering*; *AIAA SciTech Forum*, *AIAA AVIATION Forum*.

AWARDS & HONORS

2025 Korean Space Science Society Spring Conference | Best Oral Presentation Award

Apr. 2025

- Presentation Title: Progress of space debris environment and risk analysis framework development

Optimization Grand Challenge 2024 | Top 10% of Participants

Oct. 2024

- Competition to develop an optimization algorithm for pickup-and-delivery problems, sponsored by LG CNS

- Team Name: J2Opt

Hanwha-KAIST Space Hub Space Grand Challenge | Bronze

Nov. 2023

- Competition for innovative space systems, technologies, and mission concepts, sponsored by Hanwha Aerospace
- Team Name: LETA (Lunar Exploration Trajectory Analytics)
- Topic: Lunar exploration trajectory design with low-thrust propulsion and multiple gravity assist
- Prize: 3,000,000 KRW (2,050 USD)

EXTRACURRICULAR EXPERIENCE

Military Service | Defense Security Command (DSC)

Apr. 2018 – Nov. 2019

- Supported educational programs for military officers in DSC, and served as a squad leader
- Received commendation from DSC School Head (Brigadier General)

Interviewer | Humans of SNU ☎

Jul. 2017 – Dec. 2017

- Interviewed diverse members of the SNU community and discovered insightful and interesting stories

Volunteer | People to People International

Mar. 2016 – Feb. 2018

- Supported underprivileged communities in Seoul
- Student chapter union president (2017); SNU chapter president (2016)

OTHER SKILLS

Programming

- Python, MATLAB, Julia, C, C++, Java for quantitative analysis, including optimization, simulation, and statistical reasoning

Languages

- Korean (native), English